

Towards Conversationally Intelligent Dialog Systems

Jennifer Smith
SRI International, Menlo Park,
California, United States
jennifer.smith@sri.com

Aaron Spaulding
AI Center, SRI International, Menlo
Park, California, United States
aaron.spaulding@sri.com

Harry Bratt
SRI International, Menlo Park,
California, United States
harry.bratt@sri.com

Dimitra Vergyri
SRI International, Menlo Park,
California, United States
dimitra.vergyri@sri.com

Girish Acharya
AI Center, SRI International, Menlo
Park, California, United States
girish.acharya@sri.com

Kristin Precoda
SRI International, Menlo Park,
California, United States
kristin.precoda@sri.com

Andreas Kathol
SRI International, Menlo Park,
California, United States
andreas.kathol@sri.com

Colleen Richey
SRI International, Menlo Park,
California, United States
colleen.richey@sri.com

ABSTRACT

Spoken dialog systems, lacking the means to address the complex phenomena of spontaneous speech and conversational dynamics, force users into a constrained mode of dialog that resembles text-based interaction more closely than spoken conversation. Turn-taking is simplified and discourse-related information is lost, as discourse markers are largely ignored and prosodic information is not captured or utilized. We hypothesize that incorporating a few of these key conversational phenomena at specific points in a dialog will reduce cognitive load in spoken human-computer interaction and expand the potential application areas of dialog systems to tasks requiring more complex interactions. In this paper, we describe our approach to adding conversational intelligence to dialog systems and our work to date validating the hypothesis that adding conversational intelligence to existing dialog systems will significantly reduce users' cognitive load.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Natural language interfaces; Human computer interaction (HCI); HCI design and evaluation methods; User studies; • **Computing methodologies** → Artificial intelligence; Natural language processing.

KEYWORDS

Conversational intelligence, Dialog systems, Spontaneous speech, conversational AI, dialogue complexity, human computer interaction

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1 INTRODUCTION

Dialog system designers simplified the problem of converting human speech into commands that a machine can understand by forcing users to interact according to conversational rules that mimic text-based interaction [1–7]. These conversational rules limit a user's turn to a complete request or command, rather than allowing the user to converse according to the naturally complex spoken turn-taking style to which they are accustomed. When humans communicate in spoken conversation, they do not simply engage in a sequence of turns consisting of complete and grammatically correct utterances [8–16]. Instead, they participate in a process of updating the common ground in which each party actively monitors themselves and the other to ensure that the information necessary for the goal of their conversation has been correctly understood. Forcing users to adapt to a different way of conversing increases the cognitive load and frustration of interacting with dialog systems [5, 17].

Dialog system design is moving towards a more naturalistic communication style [6, 18–21]. Systems like Google Duplex attempt to model human-like conversational behavior by training systems on huge amounts of in-domain data [22]. While these systems yield impressive results, their application is limited by their dependence upon access to data. Systems that fully model human speech also introduce practical and ethical issues when they are sophisticated enough—and sound real enough—that they deceive humans [23]. We can avoid these issues and still reduce the cognitive load required to interact with a dialog system by implementing smaller changes to existing technology. These improvements will increase flexibility and usability, supporting human like conversational patterns without introducing the additional issues that may arise when a user believes they are talking to a real person rather

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than a computer system with much more limited understanding capabilities.

In this paper, we describe a novel approach to augmenting existing dialog systems with more intuitive and efficient conversational capabilities we call “conversational intelligence.” Our approach leverages information embedded in prosody and discourse markers to implement modular, domain-independent sub-dialog phenomena, or microinteractions, within existing dialog systems. We believe that the addition of conversational intelligence will improve usability, decrease a user’s cognitive load, and thereby expand the utility of dialog systems beyond simple tasks into more complex application areas. We present results from a user study, part of ongoing work towards validating this hypothesis. Finally, we provide a brief update on a follow-on study and progress towards instantiating these components in a working system.

2 PROSODY AND DISCOURSE MARKERS IN DIALOG SYSTEM DESIGN

Although researchers have long recognized the utility of prosody and discourse markers in discourse functions [1–6, 16], the most widely adopted digital assistants continue to largely ignore this information¹. This is likely because simplified conversational interactions are easier to implement and the applications that drove the development of these systems (e.g. early telephone-based information systems and digital assistants) had less to gain from prosody and discourse markers compared to applications that require complex turn-taking. Utilizing prosodic information for both input and output would open new possibilities for turn-taking management and other core design issues like recovery from errors and discoverability. For example, a user may use the discourse marker “ummm” to briefly hold the floor before they pause speaking to plan the rest of their utterance. Current digital assistants will interpret the user’s pause as the end of the turn and err by proceeding as though whatever was uttered before the “ummm” constituted a complete request, grabbing the floor at the worst possible time — in the middle of the user’s thought. Additionally, these systems force users to explicitly repeat themselves rather than recognizing self-repair communicated prosodically; e.g., we observe that these systems fail when a user attempts to correct their input “I need driving directions to 556-1st Avenue” by saying “Wait, 1st Street.”

Similarly, we observe that intonation and stress are not carefully controlled in system output, which often makes longer output difficult to interpret. Incorporating prosodic control in system output would also allow systems to forgo tedious explicit grounding in favor of more elegant solutions. For example, in the face of an ambiguous request resulting in many possible responses like “What time will Starbucks open?”, a dialog system capable of producing contrastive stress could present the hours for the closest Starbucks while efficiently hinting at the presence of other Starbucks locations: “The Starbucks on *Main Street* opens at 7.”

We propose improvements to existing technology to leverage prosodic information and discourse markers in dialog system design and enable incremental implementation of new capabilities. A statistical text-to-speech (TTS) system trained with data annotated

for prosodic changes in intonation and stress will enable systems to communicate beyond the words by controlling output prosody. An automatic speech recognition (ASR) module trained to recognize and differentiate between various discourse markers, e.g., “uh-huh” vs. “uhh,” will enable systems to incorporate new capabilities like user backchanneling to control the flow of conversation. An end-pointing module that uses multiple prosodic cues to assess when a user has finished a turn will improve system accuracy in determining the end of a turn, enabling more human-like turn-taking. Prior to incorporating these new components into a running system, we tested our hypothesis that adding specific capabilities to existing systems will reduce the cognitive load of interacting with dialog systems.

3 EVALUATION OF CI RULES

To test our hypothesis, we recruited participants to complete a conversational task while interacting with a researcher using two different sets of conversational rules: one set based on existing state-of-the-art (SOTA) design decisions and one set based on conversationally intelligent (CI) design decisions. Participants completed a secondary task simultaneously with the goals of both: (1) providing a measurable proxy for cognitive load and (2) raising the participants’ overall mental workload to mimic scenarios in which the extra cognitive load of conversing under the SOTA conversational rules would pose a strain on participants. The task design parallels real-world use cases in which people engage in a conversation with a dialog system while engaged in another task, like driving a car in stop-and-go traffic. In addition to testing our hypothesis, the study also aims to capture subjective impressions of the rule sets and validate a protocol that could be used for a future Wizard-of-Oz study to evaluate a fully automated system.

3.1 Protocol Design

Pilot runs of a protocol in which participants were told they were interacting with a dialog system rather than a human researcher revealed that most participants had a mental model of spoken human-computer interaction based on previous exposure to systems like Apple’s Siri or Amazon’s Alexa, and as a result they tended to use a limited interaction style when conversing with either a simulated SOTA or CI-enabled system. Instructing participants to engage differently without biasing them towards the CI-enabled system proved a challenge. In order to prime participants to suspend their assumptions about spoken interaction with a dialog system, we chose not to implement a Wizard-of-Oz study which requires deceiving participants into believing they were interacting with a system; instead, we instructed participants to converse with a human researcher who would use different “conversational rules” that may be implemented in future dialog systems. Thus, participants knew that the conversational rules pertained to dialog systems, but also understood that they were interacting with a human rather than a system. This approach also allowed us to test our conversational rule set before it was fully implemented in a working system.

Participants completed a conversational task while interacting remotely, via voice only, with a researcher in a separate room who used two different sets of conversational rules. To minimize variation and the potential impact of researcher bias, most of what the

¹Based on analysis of Google Home, Amazon Echo (1st generation), Siri (on iPhone XR) in November 2018.

Table 1: Conversational Rules

Rule Set	Conversational Rules
Both	The researcher cannot give information about spelling (e.g., of museum names). The researcher must proceed as if a pointer is moving linearly along the list and moving to the next item only after having stated the information for the entire list item (e.g. business name and hours). The researcher can answer specific questions about the content of an individual list item since this is easy to enable with existing systems and is already enabled for some tasks with some SOTA systems. The participant cannot influence the researcher’s rate of speech.
CI	The researcher uses intonation and pauses in both system input and output (see Table 2 for a detailed example). Since requiring a “wake word” would make complex interactive tasks tedious, the researcher is always listening and will stop speaking if the participant interrupts; this also means the participant must explicitly end the conversation. The lengthy grounding of SOTA systems is replaced with the kind of brief grounding one would expect in a human-human conversation. For example, when fulfilling a participant’s request to move to a previous museum in the list, the researcher will ground the output with an acknowledgement of the request (e.g., “Ok” or “Sure”) before reading the museum hours.
SOTA	The researcher must always “ground” system output. For example, if the participant asks about the hours for a particular museum or day, the researcher will start the system output with “The hours are” or “The Saturday hours are”. The researcher always maintains initiative; i.e., the researcher must follow each list item with a question like “Should I continue?” or after fulfilling a request to repeat the previous item in the list, the researcher asks “Would you like to hear the next museum?”. The researcher cannot use intonation or prosody when interpreting a participant’s statements or requests except when the participant uses intonation to indicate a question. The researcher stops listening and responding after a long pause and the participant must use a “wake word” to continue the conversation. Here, the SOTA rules deviate from existing SOTA systems in that the researcher does not expect the participant to deduce this rule without any explicit guidance. This type of deduction is meant to happen over time and requiring the deduction would place the SOTA system at an unfair advantage. Specifically, after the first long pause, the researcher says, “When you’re ready to continue, just say “<wake word>, and I will listen again.” If the participant speaks while the researcher is speaking, the researcher will ignore the participant and say, “I think I heard you say something while I was talking, if you want to get my attention while I’m talking, just say ‘<wake word>.’ If the participant responds with “no” to the WOZ question “should I continue?”, then WOZ will say, “When you’re ready to continue, just say “<wake word>, and I will listen again.”

researcher said was read from a script. The conversational rules attempt to govern every anticipated aspect of the conversation, including: (1) the prosody with which to read the script; (2) the researcher’s ability to understand the participant’s prosody; (3) the researcher’s ability to understand conversational behaviors like backchannelling and holding the floor; and (4) stock output phrases for particular situations (for example, how to communicate that the researcher did not understand the participant).

The main differences captured in the CI and SOTA rules relate to grounding, initiative, prosody, backchanneling, and barge-in. SOTA rules were defined by observing the output of SOTA digital assistants in response to the user requestes that generate system output in the form of a list of information². The SOTA systems always maintain initiative, ground output, ignore discourse markers, and neither produce nor understand prosodic information other than, perhaps, pitch inflections relevant to questions. In contrast, the CI rules require only minimal grounding and err on the side of giving the participant the initiative in the conversation in most circumstances. The CI rules also allow the researcher to respond to backchanneling and other discourse markers, and to consider prosodic information when interpreting the participant’s intent. Both sets of rules attempt to standardize the semantic capabilities of the researcher from conversation to conversation, varying only the conversational phenomena enabled under each set of rules.

For example, under both sets of rules, the researcher adopts the procedure of SOTA systems, reading the list output in a fixed order. Additional details about the conversational rules are documented in Table 1.

The example below in Table 2 is taken from the actual script read by the researcher and illustrates how each set of rules differs in approach to reading an individual list item. The prosodic annotation legend illustrates prosodic changes in the researcher’s speech under each set of rules. Under the CI rules, intonation functions like audible punctuation, helping the user to parse the information. For example, the rising intonation after “Monday and Tuesday” indicates that the preceding chunk of information is distinct from what comes next and also that there is another chunk of information ahead. The CI rules also prescribe pausing after a chunk of information to allow the participant to control the pace of the information, for example through a backchannel (allowing system to continue sooner) or filled pause (signaling the system should wait while the user is thinking). The researcher also uses less verbose grounding under CI rules to minimize distraction and reduce the cognitive load required to filter out extraneous information or respond to a question from the researcher like “would you like to hear more?” Specifically, after a long pause, the researcher moves

²Based on analysis of Google Home, Amazon Echo (1st generation), Siri (on iPhone XR) in November 2018.

Table 2: Example Interaction for SOTA and CI rules

Rule Set	Example Interaction
SOTA	- Following an affirmative response to the question at the end of the last list item (“<i>Should I continue?</i>”), the researcher reads: <i>“Castle Noel Museum of Art. Monday and Tuesday, closed. Wednesday to Sunday, 10AM to 2PM. Should I continue?”</i> - The researcher will continue to next item in the list after participant gives affirmative response. If the participant fails to respond or responds “No,” the researcher will instruct the participant to use the wake word when ready to continue.
CI	- Following either a backchannel, explicit command to read the next item, or sufficiently long pause, the researcher reads: <i>“There’s also Castle Noel Museum of Art^ which is closed on Monday and Tuesday* and open Wednesday to Sunday, 10AM to 2PM_”</i> - The researcher will continue to next item in the list if the participant utters a backchannel (e.g., “<i>Mmmhmm</i>”), issues an explicit command to do so, or is silent for more than 3-4 seconds. Prosodic Annotation Legend: ^ = rising intonation followed by short pause * = rising intonation with no pause _ = falling intonation followed by long pause ? = rising intonation followed by a long pause . = short pause

on to the next list item, without explicitly asking the participant for permission, unless the participant makes a request to control the flow of information. Additionally, the researcher grounds the start of the next item with a casual short phrase like “There’s also” which communicates “I’ve taken initiative and am moving on to the next item.” In contrast, under the SOTA rules, the researcher uses a flat tone throughout the list item with only short pauses to help break up the information and make it more digestible; after each list item, the researcher explicitly requests that the participant control the flow of information with the question “Should I continue?”

3.2 Procedure

The conversational task consisted of checking the accuracy of a document containing museum business hours. This task was chosen because it: (1) produces a long list of output from the system; (2) is constrained enough to limit the course of the conversation and allow the researcher to largely respond from a script; (3) is difficult enough to induce cognitive load; and (4) is consistently difficult over time to minimize any learning effect. Each participant had a unique list of museums for each of the conversational interactions; the order of the museum lists was randomized across participants, as was the order of the conversational rule sets.

To measure the cognitive load associated with each conversation, participants completed the conversational task simultaneously with a secondary computer-based detection response task designed to evaluate workload, NASA’s Multi-Attribute Task Battery (MATB) [24]. We used the System Monitoring component of MATB, as it is equivalent to a Detection Response task with the added cognitive workload of monitoring and responding to 6 channels instead of a single stimulus. To ensure that participants are aware of events even when they’re looking at the printed list of museums rather

than the computer display, a light in their peripheral vision blinks for a half second every time a MATB event fires.

The participant’s response time serves as a proxy to cognitive load as detection response tasks that are sufficiently novel (i.e., the participant is not so acclimated to the task that the reaction would be automatic) can be assumed to require cognitive control, and therefore response time will increase as cognitive load increases [25, 26]. Although cognitive load may not be additive, it’s reasonable to assume that conversing with the researcher will not lessen the cognitive load relative to performing a detection response task alone. Thus, the increase in response time when the conversation is added may be an over-estimate of the actual cognitive load required for each conversation. Any effect this has on the estimate of cognitive load will affect the estimate for each set of cognitive rules.

In addition to measuring response time, we administered surveys between conditions to capture participants’ perceived workload of each condition and subjective opinion of the system’s usability. We administered NASA’s Task Load Index (NASA-TLX), a “subjective, multidimensional assessment tool that rates perceived workload” [27]. We also administered a User Experience and Satisfaction Survey (UESS), modeled after the System Usability Scale (SUS) [28], a quick way to measure participants’ subjective opinion of the overall usability of a system. We modified the traditional wording of the SUS questions to make it grammatically appropriate for our conversational task but kept the overall structure and scoring of the typical SUS (see Table 3). Each participant also completed a survey about their use of digital assistants.

The study used a within-subjects design, where each participant was exposed to both rule sets in a single session lasting approximately 90 minutes. Participants completed a practice-run with the researcher using “human conversational rules” to prime them

Table 3: User Experience and Satisfaction Survey (UESS) Questions

1. I felt in control of the conversation	6. The interaction style seemed machine like
2. I had to think about what to say before I said it	7. I would like to talk to a computer system with this interaction style
3. The conversation flowed naturally	8. The interaction style seemed artificial
4. I felt frustrated during this conversation	9. The interaction style was polite
5. The interaction style was friendly	10. I felt the system was inflexible

towards abandoning preconceived notions about the capabilities of spoken dialog systems. They also practiced the conversational task alone each time the conversational rules changed to give them a chance to create an initial mental model for each set of conversational rules. The overall structure for each session consisted of: an introduction with general instructions; practice sessions with MATB only, conversational task only, and both tasks simultaneously; sessions with the two rule sets presented in a randomized order (first alone and then with MATB); and a debrief. Participants also completed a final round of simultaneous tasks with the human rules to measure potential learning effects. The TLX survey was administered between sessions and the UESS after the first three simultaneous tasks.

3.3 Data & Results

All twenty of the participants (11 male, 9 female) were over 18 years of age and spoke and understood English fluently. One male participant (participant 102) reported that he got frustrated halfway

through the experiment and stopped trying to accurately complete the tasks; the data from this participant is excluded from the analysis. Most (80%) of the participants reported previous experience speaking with a digital assistant, with 55% reporting using a digital assistant a few times a week or more; results did not differ significantly between those who use an assistant frequently and those who do not.

As a proxy for cognitive load, we subtracted each participant's response time to MATB when focused only on the MATB task from their response time measured while simultaneously completing the conversational task. Figure 1 shows the distribution of the difference in the proxy for cognitive load between conversations in which the researcher uses SOTA rules and CI rules. Positive values suggest that the cognitive load required to converse under the SOTA rules was longer than that of the CI rules. Overall, it took participants an average of 2.2 seconds to respond to MATB events and, on average, participants were able to respond to MATB events about 200 milliseconds (~10%) faster during conversations governed by the CI rules compared with SOTA rules. A one-sided Wilcoxon signed rank test with paired samples confirms ($P = 0.033$) that median difference in response time was significant, and greater than zero, meaning participants took longer to respond to MATB under SOTA rules. Relatively, the data show that participants responded to MATB about 23% faster in conversations governed by CI rules versus SOTA rules.

The average scores across participants for the UESS usability survey were 66.50 and 48.33 for the CI rules and SOTA rules, respectively. In general, scores below 60 can be considered as representative of low user satisfaction, which speaks to the general problems of SOTA rules for tasks with more than a minimal amount of complexity. The CI conversational rules were preferred over the SOTA rules by 14 out of the 15 participants surveyed and a one-sided paired Wilcoxon signed rank test confirmed statistical

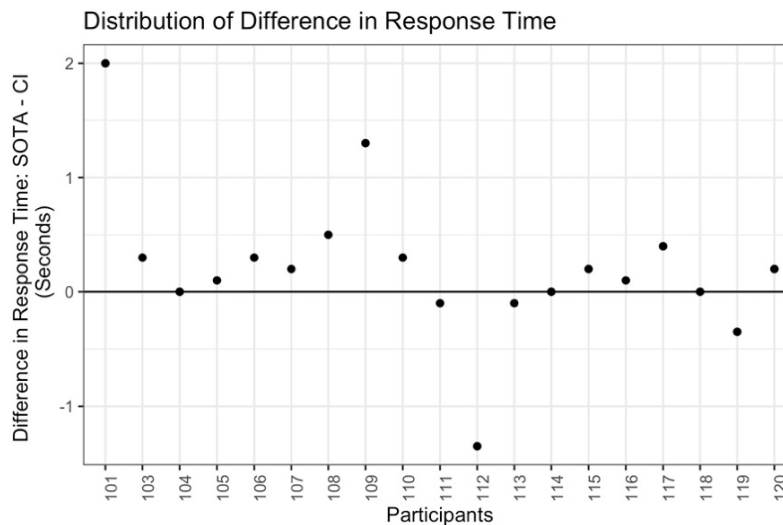


Figure 1: the distribution of the difference in the median response time to MATB between conversations in which the researcher uses SOTA rules and CI rules. Positive values mean SOTA condition median response was slower.

significance in the difference in UESS scores ($P = 0.002$). The second survey, NASA's task load survey (TLX), shows a large increase in perceived workload between the MATB-only task and the two simultaneous-task conditions, but scores for the simultaneous-tasks condition were not significantly different. Comments in the debrief with participants at the end of the study reinforce these findings: participants consistently found the simultaneous-task condition challenging and there was an overall preference for the CI system.

3.4 Discussion

The results of the user study support the hypothesis that the cognitive load induced when speaking to a dialog system can be reduced by implementing some of the design solutions outlined in this paper. Specifically, the results support our suggestion that dialog system designers adopt a few specific and achievable strategies to reduce friction by making conversations with dialog systems more like conversations between humans: reduce explicit grounding, incorporate prosodic information in system output to replace explicit grounding wherever appropriate, grant initiative to the user to facilitate control of information flow, and recognize backchannels as a means of controlling information flow.

Although participants were not interacting with a real dialog system in this study, we expect that the results may generalize to human-system conversation because, by virtue of the experimental design, the dialog was not free and fully human-like, but instead was constrained to a limited and controlled set of conversational moves. Furthermore, the participants knew that the researcher was using conversational rules designed for implementation in a future dialog system and the researcher with whom they spoke was out of sight in another room, limiting the interaction to only voice. Also, any bias introduced by the researcher in favor of the hypothesis was limited by the careful scripting of the interaction. The researcher made no significant erroneous deviations from the rules, and the deviations that were made tended to work against the hypothesis (e.g. correctly interpreting a fragment of a sentence when operating under the SOTA rules). However, further study is required to validate the generalizability as the components are realized, especially because systems in the real-world continue to be impacted by errors in understanding and latency that was not modeled in this study.

Conclusions about some of the individual conversational rules implemented in this study are not possible because they did not occur enough to measurable impact the response time to MATB. For example, we cannot draw conclusions about design decisions related to beginning and ending conversations with the system for two reasons: participants did not usually pause long enough to encounter the need to use a wake word with the SOTA system; and although participants always needed to explicitly end the conversation with the CI system, this would not greatly increase the response time to the distractor task under the CI rules because participants only needed to explicitly end the conversation in the last second of the conversational task.

4 ONGOING WORK & NEXT STEPS

We plan to expand upon the results of this study by conducting Wizard-of-Oz studies testing the effect of additional microinteractions and technology components as they are developed.

Since the study was conducted, we implemented several of our CI rules and integrated these into a live demonstration system; we trained a prosodic-markup-enhanced TTS system, a prosody-based end-pointer, and retrained our ASR system to maintain distinctions between discourse markers such as “uh” and “uh-huh”. Our TTS system was trained using Coqui TTS (see <https://github.com/coqui-ai/TTS>) from a proprietary corpus consisting of approximately 19 hours of speech recorded by a professional voice actor, along with the associated text augmented for training with novel punctuation-like symbols to represent specific prosodic phenomena. Our prosodic end-pointer uses a Convolutional Neural Network based hybrid lexical/acoustic model trained on the “self-annotated” Air Travel Information System end-pointing data set [29, 30]. We utilize a flexible microservices-based architecture, configured in a hub/spoke topology, with a new rule-based “CI Manager” as the hub. Rather than running the components through a serial pipeline, this architecture allows the manager to gather information from each of the other components to determine how to direct the dialog flow, including when to enter into and exit out of any of its rule-based microinteractions.

In future studies we plan to measure the effect on cognitive load and user satisfaction when the fully implemented components and microinteractions are employed in the interactions with the user. We are currently completing a second study in which we explore the extent to which adding prosodic cues to the system output impacts user recall, response time, and cognitive load. In this study, participants listen to a list of items generated with TTS from either a state-of-the-art system, or our system which uses CI rules to change intonation and contrastive stress. Participants are asked to recall a detail from the list, and we measure the accuracy and delay in their response, cognitive load via MATB, perceived workload via the TLX survey, and system preference. We plan to present the results of this study in a future paper.

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